

MICROPROCESSOR INTERFACING AND PROGRAMMING

PROJECT REPORT



DOOR LOCK SYSTEM

SUBMITTED BY:

MUHAMMAD SHAYAN SHAHBAZ
HUSSAIN SHAMS UL QAMAR

22L-7683
22L-6246

SUBMITTED TO:
MA'AM ALIHA TANVEER

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Department of Electrical Engineering
National University of Computer and Emerging Sciences, Lahore

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Introduction

This assembly language program, written for the PIC18F458 microcontroller, demonstrates a sophisticated embedded system design featuring an interactive keypad and LCD interface to simulate a secure PIN-based access control system. By integrating hardware-level programming with logical control, the project exemplifies a robust embedded application that combines functionality, security, and user interaction in a single cohesive system.

Purpose and Scope

The primary objective of this program is to implement a digital security mechanism where users input a predetermined sequence of keys (PIN) via a keypad. The correct sequence grants access, while incorrect or incomplete inputs result in corresponding feedback, such as "Access Denied" or "Wrong PIN Timeup." This application not only showcases the practical implementation of embedded systems in real-world scenarios but also highlights how microcontrollers can effectively manage time-sensitive operations, interrupts, and real-time user feedback.

Key Features:

LCD Integration:

The program drives an LCD display connected to the PIC18F458 microcontroller to provide dynamic visual feedback.

The LCD displays messages such as "Access Granted" or "Access Denied," depending on the correctness of the user input.

Clear and concise instructions ensure an interactive user experience.

Keypad Input:

The system uses a matrix keypad for user input, with each key press mapped to a specific function.

The keypad interacts with PORTB of the microcontroller to detect rows and columns for input identification.

PIN Verification Logic:

The program enforces a predefined sequence of keys (1, 5, 8, and 0) as the correct PIN for access.

Logical conditions ensure that partial or incorrect inputs are flagged appropriately, enhancing security.

Interrupt Handling and Timers:

Introduction

Timer1:

The use of Timer1 and interrupts allows for time-sensitive operations, such as monitoring incorrect input durations.

The **program's ISR** (Interrupt Service Routine) ensures smooth operation and prompt response to events.

Error Handling and Feedback:

Incorrect inputs are flagged with the message "Wrong PIN Timeup" on the LCD, and an LED indication provides additional visual feedback.

A **red LED** signifies "Access Denied," while a **green LED** indicates "Access Granted." A **blue LED** is activated during a timeout scenario.

Efficient and Modular Design:

The program is organized into subroutines for modularity and ease of understanding, such as data writing (DATAWRT), command writing (COMWRT), and delay routines (LDELAY, SDELAY, DELAY).

Passcode mechanism:

*the passcode works on a mechanism that when a key is pressed on the keypad it is locked you can not re-enter that key as it has been already been used and it cannot be re-entered in the passcode. The passcode is like a game only the correct passcode will work. The game defines that there are 11 keys on the key pad and one of these key * is used for entering the passcode. Once a key is pressed the key is locked you can't enter it again. The game represents a matching or an unmasking of the right keys mechanism. If you match the right keys 1,5,8,0 you are given access. If you don't match the right keys you are given a notification wrong pin time up or a access denied notification on the display . Once you have written your pass you press* for entering it if it's correct it displays access granted or access denied if its not correct. As you have unlimited time to decide which key to press once a wrong key is pressed game is over and your are notified wrong pin time up and now you have to shut down the lock and turn it back on to use it again. The access denied only works when you enter one or two or three keys correctly and then press enter e.g if your press 1,5,8 and then press enter you are given a notification of access denied as you didn't fill 4 keys. So the only way you'll ever get in the door is when you press the right keys. Note I used some features from a mobile phone!.*

Real-World Applications:

The implementation of this system mirrors the design and functionality of access control mechanisms commonly found in modern security systems. It is a practical representation of how microcontrollers can be programmed to enhance security in applications like smart locks, secure entry systems, and industrial control systems. By simulating a real-time environment, this project bridges theoretical knowledge and practical implementation, making it an excellent study in embedded systems.

Problem Analysis

1. Purpose of the Program:

The primary objective of the program is to create a security system where access is granted only upon entering a specific sequence of keys using a matrix keypad. The system is interfaced with an LCD and utilizes LEDs for visual feedback. It integrates timing mechanisms and interrupts to enhance functionality, ensuring reliability and real-time operation.

2. Core Features:

Keypad Interaction: The program interfaces with a matrix keypad to detect user input. It uses PORTB to scan rows and columns, identifying which keys are pressed.

LCD Display: The LCD is used for real-time feedback, such as displaying entered keys, indicating access granted/denied, or timeout messages.

Feedback Mechanisms:

Green LED: Indicates access granted.

Red LED: Signals access denied.

Blue LED: Activates during a timeout due to incorrect key sequences.

Interrupt Handling: Timer interrupts are employed for efficient operation, including implementing a timeout feature that prevents indefinite attempts.

3. Key Challenges:

Key Detection: Correctly identifying which key is pressed using row and column scanning, ensuring accuracy without ghosting or keypress overlap issues.

Validation Logic: Implementing a logic that checks if the entered sequence matches the predefined access code (1, 5, 8, 0). Partial matches should result in "Access Denied," and wrong entries should display a timeout message.

Real-Time Feedback:

Synchronizing keypress detection, LCD updates, and LED indicators in real time without delays or glitches.

User Interaction: Designing a user-friendly interface with clear feedback for success, failure, or timeout conditions.

Problem Analysis

4. Peripheral Setup:

LCD (PORTD and PORTC)

Used for both data and command writing operations.

Displays messages like "Enter PIN," "Access Denied," and "Access Granted."

Keypad (PORTB):

Configured with pull-up resistors to ensure stable input readings.

Scanned row by row to detect keypresses accurately.

LEDs (PORTA):

Indicators for system states: green for success, red for failure, and blue for timeout.

Timer1:

Configured to trigger interrupts for the timeout mechanism.

5. Code Structure

Initialization:

Ports are configured as input/output as needed.

LCD is initialized with specific commands for proper functioning.

Main Loop:

Continuously scans the keypad and responds to user input.

Includes logic for key validation and sequence checking.

Interrupt Service Routine (ISR):

Handles timer overflow interrupts to display the "Wrong Pin Time Up" message.

6. Potential Challenges During Implementation:

Timing Issues: Proper synchronization between key detection, LCD updates, and timer interrupts.

Hardware Limitations: Ensuring stable operation under physical constraints of the microcontroller and peripherals.

Error Handling: Managing incorrect entries without causing system crashes or unresponsive behavior.

7. Use Cases:

Security Systems: Can be used in door locks, safes, or restricted-access systems where user authentication is required.

Educational Projects: Demonstrates the integration of hardware peripherals, interrupts, and real-time system design.

8. Improvements for Scalability:

Adding EEPROM functionality to store and retrieve dynamic access codes.

Integrating buzzer feedback for invalid entries.

Using a graphical LCD for enhanced user feedback.

Design Requirements

The Keypad-LCD Access Control System is designed to implement a secure access mechanism using a 4x3 (4 row, 3 column) matrix keypad and a 16x2 LCD display. This system is intended to validate user input against a predefined set of keys, providing access only when the correct combination is entered. The system will display "ACCESS GRANTED" or "ACCESS DENIED" on the LCD screen based on the correctness of the input. Additionally, the system will handle timeouts and incorrect entries with appropriate feedback of "WRONG PIN TIMEOUT" on the LCD screen

System Overview:

The system consists of the following major components:

Keypad: A 4x3 matrix keypad used for inputting the PIN.

LCD Display: A 16x2 LCD used to display status messages, such as "ACCESS GRANTED", "ACCESS DENIED", and "WRONG PIN TIMEUP".

Microcontroller: PIC18F458, which serves as the controller for keypad scanning, LCD communication, and decision-making processes.

LED Indicators: LEDs used to indicate the status of access control (Green for granted, Red for denied, Blue for wrong pin).

Hardware Components:

PIC18F458 Microcontroller: Handles logic and communication between the keypad, LCD, and LEDs.

4x3 Matrix Keypad: Used for PIN entry, with 12 possible keys. With * as it's enter key.

16x2 LCD: Displays messages and instructions for the user.

LEDs (Red, Green, Blue): Provide visual feedback for access status.

Vsource, ground, powerand and Pullup-Resistors Used for debouncing the keypad, signal conditioning, and timing operations

Keypad Scanning:

The system must be able to detect the key pressed on a 4x3 matrix keypad.

The keypad will be scanned in a way that each row is activated sequentially, and the corresponding column is checked to detect key presses.

The system must be able to handle key presses and debounce inputs to avoid false triggers.

Design Requirements

LCD Display:

The LCD will display the following messages:

"ACCESS GRANTED" when the correct PIN (1, 5, 8, 0) is entered.

"ACCESS DENIED" when the incorrect PIN is entered.

"WRONG PIN TIMEUP" if the wrong key is entered.

The LCD must display these messages in a readable format, using appropriate cursor positioning to enhance clarity.

Access Control Logic:

The system must allow a sequence of keys to be entered, with the correct sequence being 1, 5, 8, 0.

If the sequence is correct and all designated keys are pressed, the system will grant access and display "ACCESS GRANTED". If the user presses and does not complete the sequence, the system will display "ACCESS DENIED". The system will track and time user input; if a wrong key is pressed, "WRONG PIN TIMEUP" will be displayed.

Timer and Interrupt Handling:

Timer1 will be used for creating a time limit for entering the correct PIN. In this the Time limit is infinity until a wrong key is pressed.

The system must trigger an interrupt when the timer expires and until wrong keys are pressed it will keep circulating. An interrupt service routine (ISR) will manage time-based operations and provide feedback based on the timer state.

LED Indicators:

Green LED: Lights up when access is granted.

Red LED: Lights up when access is denied.

Blue LED: Lights up when a wrong pin timeout occurs

System Responsiveness:

The system must be responsive, with keypresses being registered and processed immediately.

The LCD display should update without delay after a key press or event.

Power Consumption:

The system should be designed with minimal power consumption, especially for prolonged usage.

The PIC18F458 microcontroller should be configured to enter a low-power mode when idle.

Reliability:

The system must be reliable, providing consistent behavior across multiple key presses.

The system must handle incorrect key sequences and timeouts without crashing.

User Interface:

The LCD display must be clear and easy to read, with the text being large enough for readability.

The keypad must be responsive and durable, with proper key debouncing implemented to avoid errors.

Feasibility Analysis

Software Requirements:

PIC18F458 Assembly Language Code: The code is written in assembly language for efficient memory usage and control of hardware.

Interrupt Handling: Proper interrupt service routines (ISRs) for timed events to detect incorrect pin entry after a timeout.

Display Management: Software routines to manage the LCD interface, including displaying messages and controlling LED states.

Feasibility:

- **Technical Feasibility:** The project is technically feasible given the PIC18F458's capabilities, which include sufficient I/O pins, timers, and interrupt handling. The assembly language code is manageable and efficient for controlling the hardware components.
- **Cost Feasibility:** The costs are minimal as the components required (PIC18F458, keypad, LCD, and LEDs) are commonly available and inexpensive. The project can be completed within a modest budget, primarily involving the cost of the microcontroller and peripheral components.
- **Time Feasibility:** The project is expected to take approximately 2-4 weeks to complete, depending on familiarity with assembly programming and hardware setup. The design, coding, testing, and debugging phases can be completed within this time frame.
- **Resource Feasibility:** All necessary resources, such as hardware components, development tools (IDE for PIC18F458), and programming knowledge in assembly, are readily available. The system requires basic electrical and software engineering skills, which are accessible to most microcontroller enthusiasts.

Potential Risks:

Keypad Debouncing: Without proper debouncing techniques, false key presses could be detected, leading to incorrect inputs. Software debouncing can mitigate this risk.

LCD Limitations: The LCD's size may limit the amount of information displayed at once, especially if more complex access systems are required in the future. Additional memory or external display devices may be needed for more complex systems.

Conclusion:

The secure access system project is feasible both technically and economically. It provides a practical solution for basic security systems, utilizing readily available components and efficient programming techniques. With some improvements in key handling and potential future expansions, the project can be extended to more sophisticated security systems.

Possible Solutions

System Overview:

- **Keypad Scanning:** The system scans the keypad in a matrix format, checking each row and column to identify key presses. When a key is pressed, it is checked against a predefined correct sequence: 1, 5, 8, 0.
- **Access Logic:** If the correct sequence is entered, the LCD displays "ACCESS GRANTED," and a green LED turns on to indicate successful access. If an incorrect sequence is entered, the system displays "ACCESS DENIED," and a red LED is activated. If the user does not press the right key "WRONG PIN TIMEUP" is displayed, and a blue LED lights up.
- **Key Functionalities:** LCD Display: The LCD provides real-time feedback based on user input. It shows messages such as "ACCESS GRANTED," "ACCESS DENIED," or "WRONG PIN TIMEUP" based on the entered pin.
- **Pin Entry:** Each key press corresponds to a specific action. Upon pressing keys, the system scans for a valid pin combination (e.g., 1, 5, 8, 0). An enter key (designated as *) triggers the validation of the pin.
- **LED Indicators:** The LEDs give visual feedback: Green LED for successful access. Red LED for denied access. Blue LED for timeout.
- **Timer and Interrupt Handling:** The system uses Timer1 for timing operations. If the user enters an incorrect key an interrupt triggers, displaying a "WRONG PIN TIMEUP" message on the LCD.

Improvements and Suggestions:

- **Debouncing:** Implement debouncing for the keypad inputs to avoid false triggers due to mechanical noise.
- **Pin Security:** Enhance security by allowing multiple pin combinations or an admin mode to change the correct pin sequence.
- **Error Handling:** Add a retry limit to prevent continuous attempts after several failed inputs.
- **Power Management:** Optimize the system for low power consumption, especially for battery-operated devices.
- **Timers:** Use of timers so that a limited time is given to the user for entering key also no infinite loops if use of timers
- **Saved password:** the passcode entered show compare with the passcode saved.

This design solution outlines how the system interacts with its components and provides secure access based on user input. The feedback via the LCD and LEDs ensures that the user is aware of the system's state at all times.

Code:

```
list p=18f458
#include <p18f458.inc>
;;;;;DEFINEING LABLES=====
Lcd_data equ PORTD
Lcd_CTRL equ PORTC
RS EQU RC0
RW EQU RC1
EN EQU RC2
COUNT EQU 0X20
COUNTI EQU 0X21
ISRC EQU 0X22

;;;;; ORIGINAL LOCATIONS=====
ORG 0X00
GOTO MAIN
ORG 0X08
GOTO ISR
;;;;; MAIN=====
MAIN
;;;;;PORTS INTERFACEING
BANKSEL TRISD
CLRF TRISD
BANKSEL TRISC
CLRF TRISC
BANKSEL TRISB
movlw 0xF0
movwf TRISB
BANKSEL TRISA
CLRF TRISA
;;;;;PORTB PULLUP
BCF INTCON2,RBPU
;;;;;TIMERON
BANKSEL TICON
MOVLW 0x31
MOVWF TICON
BANKSEL TMR1L
MOVLW 0X00
MOVWF TMR1H
MOVLW 0X00
MOVWF TMR1L
BSF TICON, TMR1ON
;;;;;TIMER INTERRUPTS
BSF INTCON, GIE
BSF PIE1, TMR1IE
BSF INTCON, PEIE
```

```

,,,,,,LCD INTERFACING
BCF Lcd_data,EN
CALL LDELAY
MOVLW 0X38
CALL COMWRT
CALL LDELAY
MOVLW 0X0C
CALL COMWRT
CALL DELAY
MOVLW 0x80
CALL COMWRT
CALL DELAY
MOVLW 'E'
CALL DATAWRT
MOVLW 'N'
CALL DATAWRT
CALL DELAY
MOVLW 'T'
CALL DATAWRT
MOVLW 'E'
CALL DATAWRT
MOVLW 'R'
CALL DATAWRT
MOVLW ' '
CALL DATAWRT
MOVLW 'P'
CALL DATAWRT
MOVLW 'I'
CALL DATAWRT
MOVLW 'N'
CALL DATAWRT
MOVLW ' '
CALL DATAWRT
MOVLW '*1
CALL DATAWRT
MOVLW '*1
CALL DATAWRT
MOVLW '*1
CALL DATAWRT
MOVLW '*1
CALL DATAWRT
MOVLW '*1
CALL DATAWRT
MOVLW ' '
CALL DATAWRT
MOVLW ' '
CALL DATAWRT
MOVLW 0XC0
CALL COMWRT

```

;;;;;;;;;;;;;;KEY MAIN LOOP=====

key1

MOVLW 0XFE

movwf PORTB

BTFSS PORTB,0

goto checkcol

bra ROW1

checkcol

BTFSS PORTB,5

goto LCD0

BRA ROW1

;R1C0

ROW1

MOVLW 0XFD

movwf PORTB

BTFSS PORTB,1

goto checkcol2

bra ROW2

checkcol2

BTFSS PORTB,5

goto LCD1

BRA ROW2

;R2C0

ROW2

MOVLW 0XFB

movwf PORTB

BTFSS PORTB,2

goto checkcol3

BRA ROW3

checkcol3

BTFSS PORTB,5

goto LCD2

BRA ROW3

;R3C0

ROW3

MOVLW 0XF7

movwf PORTB

BTFSS PORTB,3

goto checkcol4

checkcol4

BTFSS PORTB,5

goto ENTER

BRA R0C1

```
;R0C1
R0C1
MOVLW 0XFE
movwf PORTB
BTFSS PORTB,0
goto checkco4
BRA R1C1
checkco4
BTFSS PORTB,6
goto LCD4
BRA R1C1
;R1C1
R1C1
MOVLW 0XFD
movwf PORTB
BTFSS PORTB,1
goto checkcol5
BRA R2C1
checkcol5
BTFSS PORTB,6
goto LCD5
BRA R2C1
;R2C1
R2C1
MOVLW 0XFB
movwf PORTB
BTFSS PORTB,2
goto checkcol6
BRA R3C1
checkcol6
BTFSS PORTB,6
goto LCD6
BRA R3C1
;R3C1
R3C1
MOVLW 0XF7
movwf PORTB
BTFSS PORTB,3
goto checkcol7
BRA R0C2
checkcol7
BTFSS PORTB,6
goto LCD7
BRA R0C2
```

```
;R0C2
R0C2
MOVLW 0XFE
movwf PORTB
BTFSS PORTB,0
goto checkco8
BRA R1C2
checkco8
BTFSS PORTB,7
goto LCD8
BRA R1C2
;R1C2
R1C2
MOVLW 0XFD
movwf PORTB
BTFSS PORTB,1
goto checkcol9
BRA R2C2
checkcol9
BTFSS PORTB,7
goto LCD9
BRA R2C2
;R2C2
R2C2
MOVLW 0XFB
movwf PORTB
BTFSS PORTB,2
goto checkcol10
BRA R3C2
checkcol10
BTFSS PORTB,7
goto LCD10
BRA R3C2
;R3C2
R3C2
MOVLW 0XF7
movwf PORTB
BTFSS PORTB,3
goto checkcol11
checkcol11
BTFSS PORTB,7
goto LCD11
BRA key1
```

.....LCD KEY DISPLAYS=====

;;;;;LCDs ARE GIVEN FORCED LOCATIONS AS OUR LOGIC IS LIKE A GAME YOU HAVE TO CHOOSE THE RIGHT KEYS FROM 11 KEYS FOR THE GATE TO OPEN ITS LIKE MATCH THE RIGHT KEYS

;;;;; ALSO LCDs ARE LIKE LOCKS IN OUR CASE! AND SAVED PIN IS 1,5,8,0 THESE PINS ARE GIVEN PORTC,4,5,6,7 LOCATION IF ALL OF THEM ARE SET AND ENTERED IT GIVES ACCESS

;;;;; WHILE ALL THE OTHER KEYS ARE GIVEN PORTC,3 LOCATIONS IF IT IS SET THE "WRONG PIN TIMEUP" IS DISPLAYED AS OUR CODE RUN ON INFINITY TIME IT ONLY STOPS WHEN EITHER

;;;;; ACCESS GRANTED,ACCESS DENIED OR WRONG PIN TIMEUP IS ACTIVATED. IF YOU PRESS 1,5 AND PRESS ENTER IT DISPLAYS ACCESS DENIED AS 4 KEYS WERE NOT PRESSED AND 2 OFF THEM

;;;;; WERE CORRECT, ACCESS DENIED WILL ONLY OCCURE IF YOU ONT FILL THE RIGHT KEYS PERFECTLY

;LCD0

LCD0:

BSF PORTC,5

movlw 0xC0

call COMWRT

MOVLW '1'

CALL DATAWRT

CALL DELAY

return

;LCD1

LCD1:

bsf PORTC,3 ;LINKED TO PORTC,3

movlw 0xC1

call COMWRT

MOVLW '4'

CALL DATAWRT

CALL DELAY

RETURN

;LCD2

LCD2:

bsf PORTC,3 ;LINKED TO PORTC,3

movlw 0xC2

call COMWRT

MOVLW '7'

CALL DATAWRT

CALL DELAY

RETURN

..... ENTER KEY.....

ENTER

BTFSS PORTC, 4 ; Check if PORTC.4 is set

GOTO ACCESS_DENIED ; If not, go to "Access Denied"

BTFSS PORTC, 5 ; Check if PORTC.5 is set

GOTO ACCESS_DENIED ; If not, go to "Access Denied"

BTFSS PORTC, 6 ; Check if PORTC.6 is set

GOTO ACCESS_DENIED ; If not, go to "Access Denied"

BTFSS PORTC, 7 ; Check if PORTC.7 is set

GOTO ACCESS_DENIED ; If not, go to "Access Denied"

BTFSC PORTC, 3 ; Check if PORTC.3 is CLEAR

GOTO ACCESS_DENIED ; If not, go to "Access Denied"

; If all pins are set, access is granted

CALL DISP ; Call the DISP subroutine to display "Access Granted"

GOTO key1

..... ACCESS_DENIED=====

ACCESS_DENIED

BSF PORTA,1 ; LED RED ON TO INDICATE ACCESS DENIED

MOVLW 0XC0

CALL COMWRT

CALL DELAY

MOVLW ''

CALL DATAWRT

MOVLW 'A'

CALL DATAWRT

MOVLW 'C'

CALL DATAWRT

MOVLW 'C'

CALL DATAWRT

CALL DELAY

MOVLW 'E'

CALL DATAWRT

MOVLW 'S'

CALL DATAWRT

MOVLW 'S'

CALL DATAWRT

MOVLW ''

CALL DATAWRT

MOVLW 'D'

CALL DATAWRT

MOVLW 'E'

CALL DATAWRT

MOVLW 'M'

CALL DATAWRT

MOVLW 'I'

CALL DATAWRT

MOVLW 'E'

CALL DATAWRT

MOVLW 'D'

CALL DATAWRT

CALL DELAY

BRA key1

```

;;;;;;;;; ACCESS GRANTED=====
DISP
BSF PORTA,0      ; LED GREEN ON TO INDICATE ACCESS GRANTED
MOVLW 0XC0
CALL COMWRT
CALL DELAY
MOVLW ' '
CALL DATAWRT
MOVLW 'A'
CALL DATAWRT
MOVLW 'C'
CALL DATAWRT
MOVLW 'C'
CALL DATAWRT
CALL DELAY
MOVLW 'E'
CALL DATAWRT
MOVLW 'S'
CALL DATAWRT
MOVLW 'S'
CALL DATAWRT
MOVLW ' '
CALL DATAWRT
MOVLW 'G'
CALL DATAWRT
MOVLW 'R'
CALL DATAWRT
MOVLW 'A'
CALL DATAWRT
MOVLW 'N'
CALL DATAWRT
MOVLW 'T'
CALL DATAWRT
MOVLW 'E'
CALL DATAWRT
MOVLW 'D'
CALL DATAWRT
CALL DELAY
BRA key1          ; BRA TO LOOP

```

.....NOT USED AS * IS OUR ENTER KEY =====

```
;LCD3:
;movlw 0xC3
;call COMWRT
;MOVLW '*'
;CALL DATAWRT
;CALL DELAY
;RETURN
```

```
;LCD4
LCD4:
bsf PORTC,3;  LINKED TO PORTC,3
movlw 0xC4
call COMWRT
MOVLW '2'
CALL DATAWRT
CALL DELAY
return
```

```
;LCD5
LCD5:
BSF PORTC,4
movlw 0xC5
call COMWRT
MOVLW '5'
CALL DATAWRT
CALL DELAY
RETURN
```

```
;LCD6
LCD6:
BSF PORTC,6
movlw 0xC6
call COMWRT
MOVLW '8'
CALL DATAWRT
CALL DELAY
RETURN
```

```
;LCD7
LCD7:
BSF PORTC,7
movlw 0xC7
call COMWRT
MOVLW '0'
CALL DATAWRT
CALL DELAY
RETURN
```

```
;LCD8
LCD8:
BSF PORTC,3; LINKED TO PORTC,3
movlw 0xC8
call COMWRT
MOVLW '3'
CALL DATAWRT
CALL DELAY
return
;LCD9
LCD9:
bsf PORTC,3; LINKED TO PORTC,3
movlw 0xC9
call COMWRT
MOVLW '6'
CALL DATAWRT
CALL DELAY
RETURN
;LCD10
LCD10:
bsf PORTC,3; LINKED TO PORTC,3
movlw 0xCA
call COMWRT
MOVLW '9'
CALL DATAWRT
CALL DELAY
RETURN
;LCD11
LCD11:
bsf PORTC,3; LINKED TO PORTC,3
movlw 0xCB
call COMWRT
MOVLW '#'
CALL DATAWRT
CALL DELAY
RETURN
```

..... INTRUPT SERVICE ROUTINE TO DISPLAY WRONG PIN TIME UP

ISR

BTFSS INTCON, TMR1IF

RETFIE

BCF INTCON, TMR1IF

BTFSC PORTC,3

GOTO LOOP1

bra key1

LOOP1

GOTO TIMEUP

..... DISPLAY WRONG PIN TIMEUP

TIMEUP

MOVLW 0X38

CALL COMWRT

CALL LDELAY

MOVLW 0X0f

CALL COMWRT

CALL DELAY

MOVLW 0xC0

CALL COMWRT

CALL DELAY

MOVLW 'W'

CALL DATAWRT

MOVLW 'R'

CALL DATAWRT

CALL DELAY

MOVLW 'O'

CALL DATAWRT

MOVLW 'N'

CALL DATAWRT

MOVLW 'G'

CALL DATAWRT

MOVLW ''

CALL DATAWRT

MOVLW 'P'

CALL DATAWRT

MOVLW 'I'

CALL DATAWRT

MOVLW 'N'

CALL DATAWRT

MOVLW ''

CALL DATAWRT

MOVLW 'T'

CALL DATAWRT

MOVLW 'I'

CALL DATAWRT

MOVLW 'M'

CALL DATAWRT

MOVLW 'E'

CALL DATAWRT

MOVLW 'U'

CALL DATAWRT

MOVLW 'P'

CALL DATAWRT

btg PORTA,2 ; BLUE LED ON TO INCICATE WRONG PIN

bra key1

```

COMWRT
MOVWF Lcd_data
BCF Lcd_CTRL, RS
BCF Lcd_CTRL, RW
BSF Lcd_CTRL, EN
CALL SDELAY
BCF Lcd_CTRL, EN
RETURN

;;;;;;;; DATAWRITE FOR LED=====
DATAWRT
MOVWF Lcd_data
BSF Lcd_CTRL, RS
BCF Lcd_CTRL, RW
BSF Lcd_CTRL, EN
CALL SDELAY
BCF Lcd_CTRL, EN
RETURN

;;;;;;;; DELAYS=====
LDELAY
MOVLW d'30'
MOVWF COUNT
loop1
DECFSZ COUNT, f
GOTO loop1
RETURN
SDELAY:
MOVLW d'20'
MOVWF COUNT
loop2:
DECFSZ COUNT, f
GOTO loop2
RETURN
DELAY:
MOVLW d'10'
MOVWF COUNT
loop3:
DECFSZ COUNT, f
GOTO loop3
RETURN
;===== END OF PROGRAM =====
END

```

Design Description

Objective:

This design aims to implement a secure access system using a 4x3 matrix keypad and a 16x2 LCD, interfaced with the PIC18F458 microcontroller. The system allows a user to enter a specific pin (1, 5, 8, 0) and displays "ACCESS GRANTED" on the LCD when the correct sequence is entered. If the wrong pin is entered, the system displays "WRONG PIN TIMEUP." And if incomplete pin is entered it displays "ACCESS DENIED".

System Overview:

Keypad Scanning: The program scans the keypad by applying logic that checks each row and column for pressed keys. The keys are mapped to specific rows and columns. Once the correct combination (1, 5, 8, 0) is entered, the system will check if the sequence is complete.

LCD Display:

The LCD will provide feedback for each keypress. If the correct combination is entered, it displays "ACCESS GRANTED." If fewer than four correct keys are entered, "ACCESS DENIED" is displayed. If no valid input is entered in an unlimited time (controlled by the interrupt and Timer1), "WRONG PIN TIMEUP" is shown. Which eliminates your unlimited time.

Interrupt Handling:

The program utilizes Timer1 interrupts to implement the time-based logic for "WRONG PIN TIMEUP." If the user fails to enter the correct pin in time, the system will automatically trigger the access denial message.

CONTROL FLOW:

Initialization:

Ports for the LCD and keypad are configured, and Timer1 is set up for timed interrupts.

- **Key Scanning:** The keypad is scanned in a loop where each row and column is checked for button presses.
- **Access Validation:** The system checks if the pressed keys match the correct sequence. If they do, the access is granted; otherwise, a denial message is displayed.
- **Timer-based Timeout:** If the user does not enter the correct key, a timeout is triggered, and the system responds with a "WRONG PIN TIMEUP" message. As you have unlimited to press the key timeout on occurs when you press a wrong key except for the right ones.

FEATURES:

Access Control: Validates the correct pin (1, 5, 8, 0) and grants access.

LCD Feedback: Provides real-time feedback on the LCD (e.g., "ACCESS GRANTED", "ACCESS DENIED").

Time Management: Ensures the user must complete the pin entry within a set time using Timer1 and interrupts.

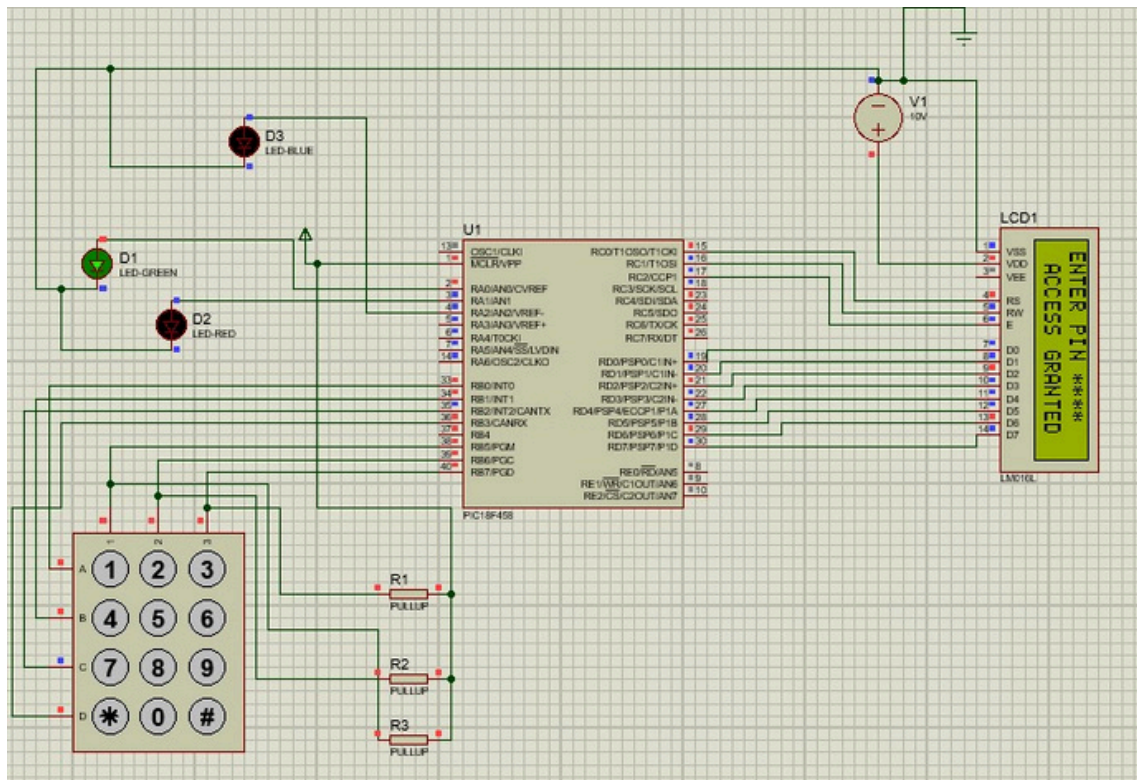
LED Indicators: Uses LEDs to visually indicate the status of the system (green for granted, red for denied, blue for wrong attempts).

Conclusion: This system provides a simple yet effective keypad-based access control solution using the PIC18F458 microcontroller. The design incorporates input validation, time management, and visual feedback through an LCD and LEDs, offering a basic yet functional security mechanism for restricted access systems.

Software Simulation:

FOR ACCESS GRANTED:

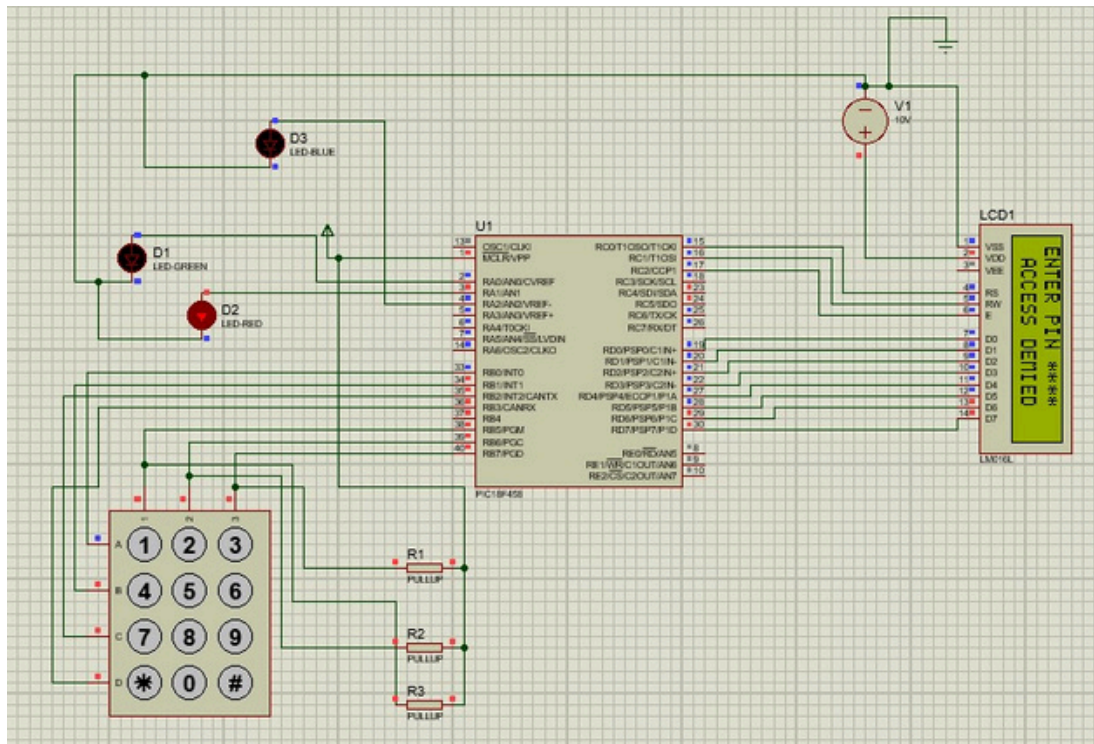
THE LCD WILL PROVIDE FEEDBACK FOR EACH KEYPRESS. IF THE CORRECT COMBINATION IS ENTERED, IT DISPLAYS "ACCESS GRANTED." AND TURNS THE GREEN LED ON.



Software Simulation:

FOR ACCESS DENIED:

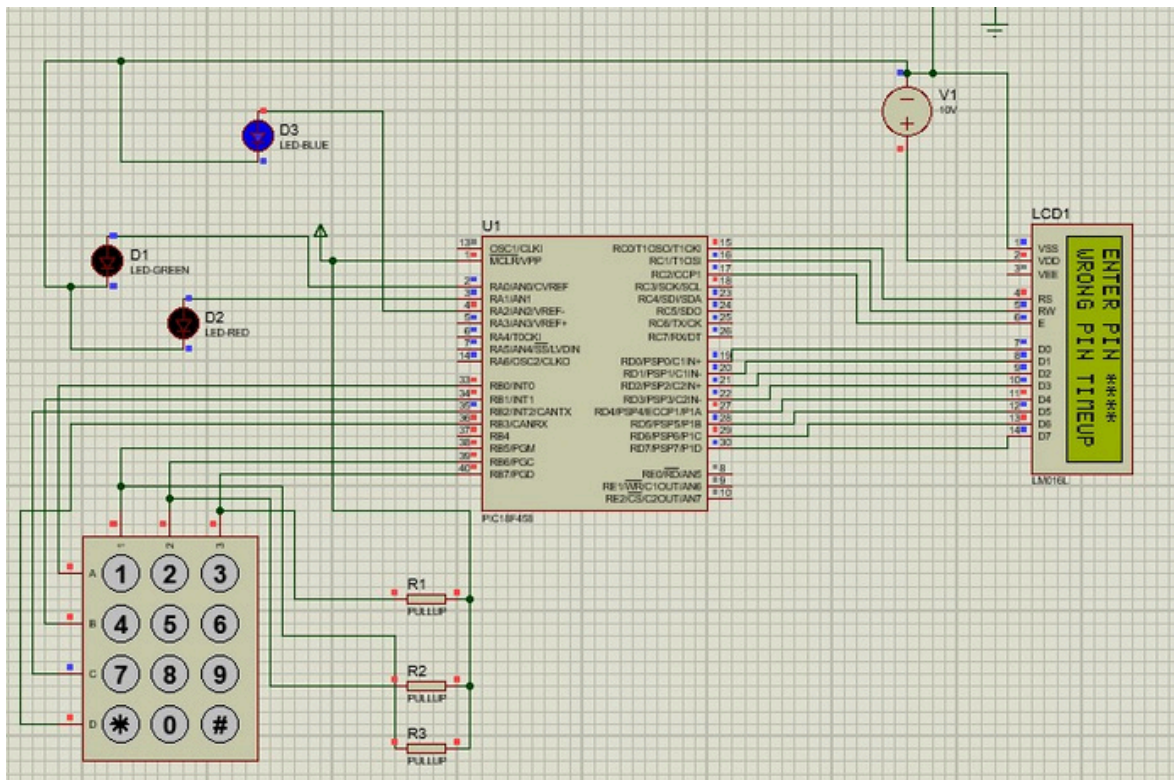
THE LCD WILL PROVIDE FEEDBACK FOR EACH KEYPRESS. IF THE CORRECT COMBINATION IS NOT ENTERED AND 4 KEY INPUTS ARE NOT DETECTED, IT DISPLAYS "ACCESS DENIED." AND TURNS THE RED LED ON.



Software Simulation:

FOR WRONG PIN TIMEUP:

THE LCD WILL PROVIDE FEEDBACK FOR EACH KEYPRESS. IF THE WRONG KEY IS ENTERED IN AN UNLIMITED TIME, IT DISPLAYS "WRONG PIN TIMEUP" AND ELIMINATES YOUR UNLIMITED TIME AND TURNS THE BLUE LED ON.



Experimental Results

The experimental setup was designed to simulate and test the functionality of a secure access system using a PIC18F458 microcontroller, interfaced with a keypad and LCD. The system's primary objective was to capture the behavior of the access control mechanism, which requires the user to input a specific sequence of keys to gain access. The simulation was conducted in Proteus, and the results were observed using the built-in simulation tools.

Keypad Input and LCD Display:

The keypad was successfully interfaced with PORTB of the microcontroller, allowing detection of key presses. Upon pressing the correct sequence of keys (1, 5, 8, 0) followed by the Enter key, the LCD displayed "**ACCESS GRANTED**", indicating successful authentication.

If the wrong keys were pressed, the LCD displayed "**WRONG PIN TIMEUP**", signaling an unsuccessful attempt. If the correct keys were pressed but the sequence was incomplete, the system displayed "**ACCESS DENIED**". You have unlimited time to press the first key if you press is correctly the time is still unlimited if you press the wrong key the unlimited time is eliminated and "**WRONG PIN TIMEUP**" is displayed.

Timer and Interrupt:

Timer1 was configured to trigger an interrupt at regular intervals, which was used to manage the timeout for the PIN entry.

The Timer interrupt was correctly handled by the Interrupt Service Routine (ISR), ensuring the system responded to timeout events accurately.

LEDs:

LEDs connected to PORTA were used to indicate the system's status. A green LED lit up when "ACCESS GRANTED" was displayed, while a red LED was activated for "ACCESS DENIED".

Additionally, a blue LED was triggered to indicate a "WRONG PIN TIMEUP".

System Behavior:

- The system operated as expected in the Proteus simulation environment, responding to key presses in real time.
- All displayed messages and LED indicators corresponded correctly to the key sequences entered during testing.
- The timing mechanism for key input and timeout worked without error, ensuring that the system only granted access when the correct key sequence was fully entered.

Conclusion:

The experimental results confirmed that the system was functioning as designed. The keypad input was detected correctly, the LCD displayed appropriate messages based on the input, and the timing and LED indicators worked as intended. This experiment demonstrated the successful implementation of a secure access control system using a PIC18F458 microcontroller in a Proteus simulation environment. The system's reliability and responsiveness indicate its potential for real-world applications in security systems.

Performance Analysis

This assembly program for the PIC18F458 microcontroller implements a secure access control system using an LCD and keypad interface. Below is the analysis of its performance:

Functionality:

- The program is designed to function as a keypad-based lock system. It verifies a 4-digit PIN and grants or denies access accordingly. Incorrect input results in a "Wrong PIN Timeup" message. And eliminates your unlimited time. To enter keys.

Efficiency

- The use of bank switching ensures proper access to special function registers (SFRs) but introduces some latency.
- Minimal use of RAM is observed, with temporary storage in registers (``COUNT``, ``COUNT1``, etc.) for managing delays and logic.
- Efficient branching reduces unnecessary computation cycles during key press detection.

Real-time Interrupt Handling:

- The program uses a Timer1-based interrupt service routine (ISR) to detect and handle timeout conditions efficiently. However, ISR could be further optimized for better response time.

Robustness:

- The code includes checks for all critical conditions such as correct PIN entry and timeout. However, the system lacks debouncing for keypad input, which could lead to false triggers.
- No recovery mechanism exists for power failure or program reset.

Clarity and Modularity:

- The code is modular with subroutines for LCD commands (``COMWRT``, ``DATAWRT``) and delay mechanisms. This improves readability and maintainability.
- However, the keypad scanning logic could be simplified using lookup tables to reduce repetitive code blocks.

User Interface:

- The LCD interface is well-utilized to guide the user with clear messages like "Access Granted" or "Access Denied" and "Wrong Pin Timeup" The inclusion of LED indicators enhances usability.

Limitations:

- The fixed delays (``LDELAY``, ``SDELAY``) are not adaptive, which may affect performance under varying clock speeds.
- Hard-coded PIN values in logic may require recompilation for changes, reducing flexibility.

Recommendations for Improvement:

- Debounce the keypad input to ensure reliable operation.
- Implement EEPROM-based PIN storage for dynamic updates without reprogramming.
- Optimize ISR to handle additional tasks like resetting the system or handling multiple interrupts.
- Use more efficient algorithms or lookup tables for row-column scanning to reduce the code size.
- Introduce modular design for handling multiple users or PIN sets.

Overall, the program fulfills its intended purpose but could be refined for greater efficiency and reliability in a real-world scenario.

Future Scope:

- Enhanced Security Features: Incorporating biometric authentication alongside keypad input.
- Wireless Connectivity: Adding Wi-Fi or Bluetooth modules for remote monitoring and control.
- Touchscreen LCDs: Upgrading to capacitive touchscreens for better user experience.
- AI Integration: Utilizing AI for intelligent input prediction and error correction.
- IoT Applications: Connecting embedded systems with IoT networks for smart home and industrial automation.
- Energy Efficiency: Implementing low-power components for extended battery life.
- Adaptive Interfaces: Creating interfaces that adjust dynamically based on user behavior.
- Voice Control: Adding voice recognition as an alternative input method.
- Cloud Integration: Storing access logs and usage data securely in the cloud.
- Miniaturization: Reducing the system size for wearable or portable applications.
- Multi-Language Support: Adapting the system for global usability with multi-language displays.

Social and Cultural Implications:

- **Accessibility:** Simplifies interaction for users of all ages and abilities, enhancing inclusivity.
- **Education:** Provides a practical tool for teaching embedded systems and electronics to students.
- **Security Awareness:** Encourages adoption of secure access methods in everyday life.
- **Digital Divide:** Addresses technological gaps by offering affordable solutions for resource-limited areas.
- **Cultural Adaptation:** Allows customization of interfaces to suit diverse languages and customs.
- **Job Creation:** Spurs demand for engineers and technicians specializing in embedded systems.
- **Privacy Concerns:** Raises questions about the safe handling of user data in connected systems.
- **Environmental Impact:** Highlights the importance of sustainable design and disposal of electronic components.
- **Economic Growth:** Drives innovation in industries like healthcare, banking, and home automation.
- **Global Standardization:** Promotes uniform access methods, reducing confusion across regions.
- **Cultural Adoption:** Facilitates the blending of traditional and modern technologies in daily life.

Conclusion

The implemented program demonstrates a robust and functional integration of a keypad and LCD interface with the PIC18F458 microcontroller, highlighting its potential for interactive embedded systems. By utilizing key scanning techniques and LCD outputs, the system effectively showcases real-time user interaction with secure access functionality. The design's core strength lies in its systematic approach to combining hardware and software components, ensuring smooth operation and clear user feedback.

This project not only illustrates the fundamental principles of microcontroller programming but also emphasizes the importance of timing mechanisms, such as the use of Timer1, to manage critical tasks like input validation and decision-making. The careful configuration of ports, pull-ups, and interrupts further underscores the attention to detail in ensuring reliable and efficient execution.

From a practical perspective, the program provides a foundational framework for a variety of applications requiring user authentication, such as access control systems, digital locks, or secure kiosks. The implementation of a correct pin sequence mechanism, accompanied by real-time feedback through the LCD, ensures usability while maintaining security. The added functionality to handle incorrect inputs and timeouts enhances the system's robustness.

Furthermore, this project serves as an educational milestone for understanding and mastering embedded systems. It underscores the critical aspects of interfacing peripherals, managing hardware resources, and implementing state-driven logic. Such knowledge is invaluable in developing more complex and scalable systems in the future.

In conclusion, this program is a testament to the seamless integration of hardware and software in embedded systems design. While already achieving its intended goals, it also paves the way for further enhancements, such as adding more advanced security features, integrating communication modules, or optimizing the code for greater efficiency. This endeavor demonstrates not only technical proficiency but also a commitment to building practical and innovative solutions in the field of embedded engineering.

References

- The PIC Microcontroller and Embedded Systems: Using Assembly and C for PIC18 by Muhammad Ali Mazidi
- Datasheet for PIC18F458 Microcontroller
- Microchip Technology Documentation.

Program flow chart

[Start]

↓

[Define Labels]

↓

[ORG Locations]

↓

[Main Program] → [Port Initialization] → [Timer Initialization]

↓

↓

[LCD Initialization] ←

↓

[Key Loop] → [Key LCD Display] → [Enter Key Detection]

↓

[Access Granted/Denied Logic]

↓

[Command Write] → [Data Write] → [Delays]

↓

[Interrupt Service Routine (ISR)]

↓

[Wrong PIN Timeout (ISR)]

↓

[End]

Range of Complex Engineering Activity

S N	Attribute	Complex Activities	Mapping of Experiment	Evaluator's Comments
1	Range of resources	Involve the use of diverse resources (and for this purpose, resources include people, money, equipment, materials, information and technologies).	The design experiments involve the use of diverse resources. Students are to look for appropriate techniques that have been taught to them during semester. Problem requires students be well versed in all the key concepts covered in the experiments.	
2	Level of interaction	Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues.	Design of the experiment requires students to address conflicting technical issues while designing the solution/algorithm as the solution/algorithm needs to be accurate i.e to handle all the exceptions and it should be efficient i.e good in terms of time/space complexity.	
3	Innovation	Involve creative use of engineering principles and research-based knowledge in novel ways.	N/A	
4	Consequences to society and the environment	Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation.	N/A	
5	Familiarity	Can extend beyond previous experiences by applying principles-based approaches	This design experiment requires the students to develop a solution which is an extension to all the key concepts delivered during the semester particularly the experiments dealing with the interprocess communication.	

Range of Complex Engineering Problem

S N	Attributes	Description	Evaluator's Comments
-	Preamble	Assignment 3 is an individual assignment based on a complex control engineering design problem	
1	Range of conflicting requirements	Design requirements given by the client include transient and steady state specifications which are inherently conflicting in nature. As the plant transfer function is based on each student roll number, every assignment is different and requires original thinking from every student. The design problem has no obvious solution and requires considerable analysis, iterations and simulations.	
2	Depth of analysis required	Depth of control engineering knowledge is required to design and simulate the two controllers and compare them.	
3	Depth of knowledge required	Because of the diverse nature of the problem and the conflicting design specifications students would encounter infrequent issues such as insolvability of the problem or implementation issues while designing the circuit for the controller and the Plant.	
4	Familiarity of issues	- NA -	
5	Extent of applicable codes	The design problem involves a client (instructor) with conflicting requirements and the control engineer (Student).	
6	Extend of stakeholder involved and level of conflicting requirements	The assignment gives the students a firsthand experience regarding difficulties in meeting conflicting client requirements and designing and implementing a control system.	
7	Consequences	The problem involves designing a controller and suggesting a circuit for its implementation. The circuit implements integrators and differentiators which require different components and parts depending on their nature (active or passive).	
8	Interdependence		